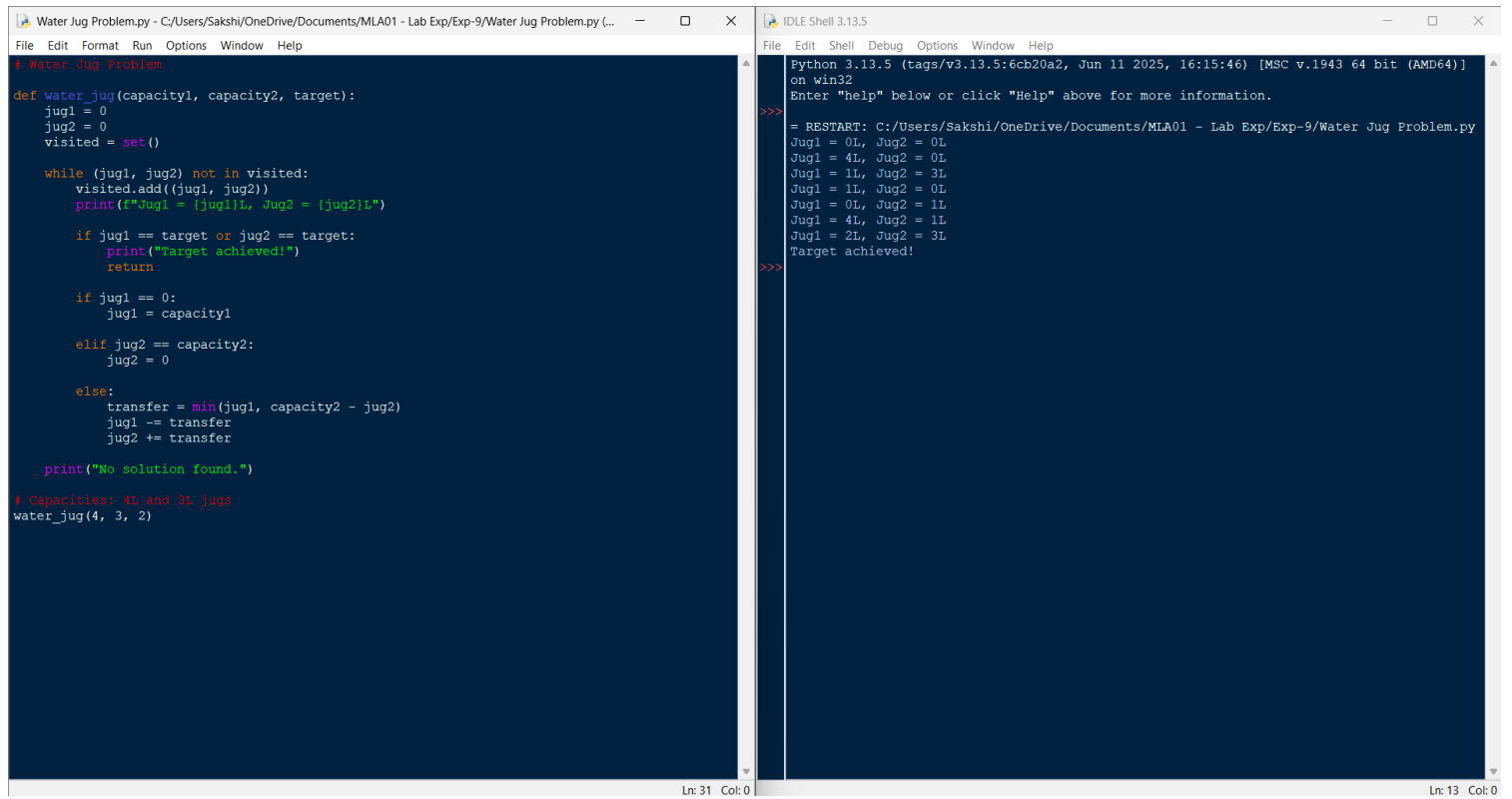


Exp:9 Water Jug Problem

Program:

```
def water_jug(capacity1, capacity2, target):  
    jug1 = 0  
    jug2 = 0  
    visited = set()  
    while (jug1, jug2) not in visited:  
        visited.add((jug1, jug2))  
        print(f'Jug1 = {jug1}L, Jug2 = {jug2}L')  
        if jug1 == target or jug2 == target:  
            print("Target achieved!")  
            return  
        if jug1 == 0:  
            jug1 = capacity1  
        elif jug2 == capacity2:  
            jug2 = 0  
        else:  
            transfer = min(jug1, capacity2 - jug2)  
            jug1 -= transfer  
            jug2 += transfer  
        print("No solution found.")  
water_jug(4, 3, 2)
```

Output:



The image shows a screenshot of a Python IDE with two windows. The left window displays the source code for a 'Water Jug Problem' solver. The code defines a function 'water_jug' that takes two jug capacities and a target amount. It uses a while loop and a set to track visited states until the target is reached or no solution is found. The right window shows the execution output, which lists the sequence of states (jug1, jug2) until the target of 4 units is achieved in jug 1.

```
Water Jug Problem.py - C:/Users/Sakshi/OneDrive/Documents/MLA01 - Lab Exp/Exp-9/Water Jug Problem.py (...)
```

```
File Edit Format Run Options Window Help
```

```
# Water Jug Problem
```

```
def water_jug(capacity1, capacity2, target):
```

```
    jug1 = 0
```

```
    jug2 = 0
```

```
    visited = set()
```

```
    while (jug1, jug2) not in visited:
```

```
        visited.add((jug1, jug2))
```

```
        print(f"Jug1 = {jug1}L, Jug2 = {jug2}L")
```

```
        if jug1 == target or jug2 == target:
```

```
            print("Target achieved!")
```

```
            return
```

```
        if jug1 == 0:
```

```
            jug1 = capacity1
```

```
        elif jug2 == capacity2:
```

```
            jug2 = 0
```

```
        else:
```

```
            transfer = min(jug1, capacity2 - jug2)
```

```
            jug1 -= transfer
```

```
            jug2 += transfer
```

```
        print("No solution found.")
```

```
# Capacities: 4L and 3L jugs
```

```
water_jug(4, 3, 2)
```

```
Ln: 31 Col: 0
```

```
IDLE Shell 3.13.5
```

```
File Edit Shell Debug Options Window Help
```

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
```

```
on win32
```

```
Enter "help" below or click "Help" above for more information.
```

```
>>>
```

```
= RESTART: C:/Users/Sakshi/OneDrive/Documents/MLA01 - Lab Exp/Exp-9/Water Jug Problem.py
```

```
Jug1 = 0L, Jug2 = 0L
```

```
Jug1 = 4L, Jug2 = 0L
```

```
Jug1 = 1L, Jug2 = 3L
```

```
Jug1 = 1L, Jug2 = 0L
```

```
Jug1 = 0L, Jug2 = 1L
```

```
Jug1 = 4L, Jug2 = 1L
```

```
Jug1 = 2L, Jug2 = 3L
```

```
Target achieved!
```

```
>>>
```

```
Ln: 13 Col: 0
```